Exercise 15.3

Solve the following equations:

1. $\frac{dy}{dx} = \frac{y - 2x}{x}$

Sol^n: The given equation can be written as $\frac{dy}{dx} = \frac{y}{x} - 2$ ... (i)

Which is homogeneous differential equation of the form $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$

let $y = vx$ $\Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$\therefore$ From (i), $v + \frac{xdv}{dx} = v - 2$

$\Rightarrow \frac{xdv}{dx} = -2$

Integrating, we get

$\int dv = \int -2\frac{dx}{x}$ $\Rightarrow v = -2\log x + c$

$\Rightarrow \frac{y}{x} = -2\log x + c$ $\Rightarrow y + 2x \log x = cx..$

2. $\mathbf{xy\frac{dy}{dx} = x^2 + y^2}$

Sol^n: The given equation can be written as $\frac{dy}{dx} = \frac{x^2 + y^2}{xy}$ ..... (i)

Which is homogeneous differential equation,

Let $y = vx$ $\Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$\therefore$ From (i), $v + x\frac{dv}{dx} = \frac{1 + v^2}{v}$ $\Rightarrow x\frac{dv}{dx} = \frac{1 + v^2}{v} - v$

$\Rightarrow \frac{xdv}{dx} = \frac{1}{v}$ $\Rightarrow vdv = \frac{dx}{x}$

Integrating, we get

$\int vdv = \int \frac{dx}{x}$ $\Rightarrow \frac{v^2}{2} = \log x + c$

$\Rightarrow \mathbf{\frac{y^2}{2x^2} = (\log x + c)}$ $\Rightarrow y^2 = 2x^2(\log x + c).$

3. $\mathbf{x^2\frac{dy}{dx} + y^2 = xy}$

Sol^n: The given equation can be written as $\frac{dy}{dx} = \frac{xy - y^2}{x^2}$ which is homogeneous.

Let $y = vx$ $\Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$\therefore$ $v + x\frac{dv}{dx} = v - v^2$ $\Rightarrow -\frac{dv}{v^2} = \frac{dx}{x}$

Integration, we get

$$\int \frac{dv}{v^2} = \int \frac{dx}{x} \qquad \Rightarrow -\left(\frac{v^{-2+1}}{-2+1}\right) = \log x + c$$

$$\Rightarrow \quad \frac{1}{v} = \log x + c \qquad \Rightarrow \frac{x}{y} = (\log x + c) \qquad \Rightarrow x = y\,(\log x + c).$$

4. $\mathbf{\frac{dy}{dx} = \frac{y}{x} + \tan\frac{y}{x}}$

Sol^n: This equation being homogeneous differential equation.

Let, $y = vx \qquad \Rightarrow \frac{dy}{dx} = v + x\frac{dy}{dx}$

$$\Rightarrow \quad v + x\frac{dv}{dx} = \frac{vx}{x} + \tan\frac{vx}{x} \Rightarrow v + x\frac{dv}{dx} = v + \tan v$$

$$\Rightarrow \quad x\frac{dv}{dx} = \tan v \qquad \Rightarrow \cot v\, dv = \frac{dx}{x}$$

$$\Rightarrow \quad \frac{\cos v}{\sin v}\, dv = \frac{dx}{x}$$

Integrating, we get

$\log \sin v = \log x + \log c \qquad \Rightarrow \log \sin v = \log cx$

$$\Rightarrow \quad \sin v = cx \Rightarrow \sin\left(\frac{y}{x}\right) = cx.$$

5. $\mathbf{x(x - y)dy = y(x + y)dx}$

Sol^n: $\frac{dy}{dx} = \frac{y(x + y)}{x(x - y)}$ ..... (i)

This equation being homogeneous differential equation.

Let, $y = vx \qquad \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

Then equation (i) becomes

$$v + x\frac{dv}{dx} = \frac{vx(x + vx)}{x(x - vx)} \qquad \Rightarrow v + x\frac{dv}{dx} = \frac{v(1 - v)}{1 - v}$$

$$\Rightarrow \quad x\frac{dv}{dx} = \frac{v(1 + v)}{1 - v} - v \qquad \Rightarrow x\frac{dv}{dx} = \frac{2v^2}{1 - v}$$

$$\Rightarrow \quad x\frac{dv}{dx} = \frac{2v^2}{1 - v} \qquad \Rightarrow \frac{1 - v}{2v^2}\, dv = \frac{dx}{x}$$

$$\Rightarrow \quad \left(\frac{1}{2v^2} - \frac{1}{2v}\right) dv = \frac{dx}{x}$$

Integrating, we get

$$\Rightarrow \quad -\frac{1}{2v} - \frac{1}{2}\log v = \log x + \log\sqrt{c}\ , \text{ where } \log\sqrt{c} \text{ is arbitrary constant.}$$

$$\Rightarrow \quad -\frac{1}{2v} = \log\sqrt{v} + \log x + \log\sqrt{c}$$

$$\Rightarrow \quad -\frac{1}{2v} = \log\left(\sqrt{vc}\ .\ x\right) \qquad \Rightarrow \frac{1}{v} = -2\log\left(\sqrt{vc}\ .\ x\right)$$

$\Rightarrow \quad \frac{1}{v} = -\log\left(\sqrt{vc} \cdot x\right)^2 \qquad \Rightarrow \frac{1}{v} = -\log(vcx^2)$ ..... (ii)

$\Rightarrow \quad \frac{x}{y} = -\log\left(\frac{y}{x} \cdot c \cdot x^2\right) \qquad \Rightarrow x = -y \log(cxy)$

$\Rightarrow \quad x + y\log(cxy) = 0.$

6. $\mathbf{(x^2 - y^2)\frac{dy}{dx} = xy.}$

Sol^n: This equation can be written as

$$\frac{dy}{dx} = \frac{xy}{x^2 - y^2} \quad \text{..... (i)}$$

Which is homogeneous differential equation

Let $y = vx$

$\therefore \quad \frac{dy}{dx} = v + x\frac{dv}{dx}$

Hence, equation (i) will becomes

$$v + x\frac{dv}{dx} = \frac{v}{1 - v^2} \qquad \Rightarrow x\frac{dv}{dx} = \frac{v}{1 - v^2} - v$$

$$\Rightarrow \quad x\frac{dv}{dx} = \frac{v - v + v^3}{1 - v^2} \qquad \Rightarrow \frac{(1 - v^2)dv}{v^3} = \frac{dx}{x}$$

$$\Rightarrow \quad \left(\frac{1}{v^3} - \frac{1}{v}\right)dv = \frac{dx}{x}$$

Integrating, we get

$$\Rightarrow \quad -\frac{1}{2v^2} - \log v = \log x + \log c \Rightarrow \frac{-1}{2v^2} = \log cxv$$

$$\Rightarrow \quad \frac{1}{v^2} = -2\log cxv \qquad \Rightarrow \frac{x^2}{y^2} = -2\log\left(cx \cdot \frac{y}{x}\right)$$

$$\Rightarrow \quad x^2 = -2y^2 \log(cy) \qquad \Rightarrow x^2 + 2y^2 \log(cy) = 0.$$

7. $\mathbf{(x^2 + y^2)\,dy = xydx}$

Sol^n: The equationcan be written as

$$\frac{dy}{dx} = \frac{xy}{x^2 + y^2} \quad \text{..... (i)}$$

This equation being homogeneous differential equation

Let $y = vx \qquad \therefore \frac{dy}{dx} = v + x\frac{dv}{dx}$

Then equation (i) becomes

$$v + x\frac{dv}{dx} = \frac{x.vx}{x^2 + (vx)^2} \qquad \Rightarrow x\frac{dv}{dx} = \frac{v - v - v^3}{1 + v^3} = \frac{-v^3}{1 + v^2}$$

$$\Rightarrow \quad \frac{1 + v^2}{v^3}dv + \frac{dx}{x} = 0 \qquad \Rightarrow \left(\frac{1}{v^3} + \frac{1}{v}\right)dv + \frac{dx}{x} = 0$$

Integrating, we get

$$-\frac{1}{2v^2} + \log v + \log x + \log c = 0 \Rightarrow -\frac{1}{2v^2} + \log vxc = 0$$

$$\Rightarrow \quad \frac{1}{v^2} = 2\log vcx \qquad \Rightarrow \left(\frac{x}{y}\right)^2 = 2\log\left(\frac{y}{x}x \cdot c\right)$$

$$\Rightarrow \quad x^2 = 2y^2 \log(cy).$$

8. $\frac{dy}{dx} = \frac{x^2 + y^2}{2x^2}$

Sol^n: $\frac{dy}{dx} = \frac{x^2 + y^2}{2x^2}$ ..... (i)

This equation being homogeneous,

Let, $y = vx$ $\Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

Hence, equation (i) will be

$v + x\frac{dv}{dx} = \frac{x^2 + v^2x^2}{2x^2}$ $\Rightarrow v + x\frac{dv}{dx} = \frac{1 + v^2}{2}$

$\Rightarrow \quad x\frac{dv}{dx} = \frac{1 + v^2}{2} - v$ $\Rightarrow x\frac{dv}{dx} = \frac{1 + v^2 - 2v}{2}$

$\Rightarrow \quad x\frac{dv}{dx} = \frac{(1 - v)^2}{2}$ $\Rightarrow \frac{dv}{(1 - v)^2} = \frac{1}{2}\frac{dx}{x}$

Integrating, we get

$\int \frac{dv}{1 - v} = \frac{1}{2}\log x + \frac{1}{2}\log c$, where $\frac{1}{2}\log c$ is constant.

$\Rightarrow \quad \frac{1}{1 - v} = \frac{1}{2}\log(xc)$ $\Rightarrow 2 = (1 - v)\log(cx)$

$\Rightarrow \quad 2 = \left(1 - \frac{y}{x}\right)\log(cx)$ $\Rightarrow 2x = (x - y)\log(cx)$.

9. $\mathbf{xdy - ydx = \sqrt{x^2 + y^2}\,dx}$

Sol^n: The equation can be rewrite as

$\frac{dy}{dx} = \frac{y}{x} + \frac{\sqrt{x^2 + y^2}}{x}$ ..... (i)

This equation being homogeneous differential equation,

Let $y = vx$ $\Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

Then from equation (i),

$v + x\frac{dv}{dx} = \frac{vx}{x} + \frac{\sqrt{x^2 + v^2x^2}}{x}$ $\Rightarrow v + v\frac{dv}{dx} = v + \sqrt{1 + v^2}$

$\Rightarrow \quad x\frac{dv}{dx} = \sqrt{1 + v^2}$ $\Rightarrow \frac{dv}{\sqrt{1 + v^2}} = \frac{dx}{x}$

Integrating, we get

$\log(v + \sqrt{1 + v^2}) = \log x + \log c$

$\Rightarrow \quad v + \sqrt{1 + v^2} = cx$

$\Rightarrow \quad \frac{y}{x} + \sqrt{1 + \frac{y^2}{x^2}} = cx$

$\Rightarrow \quad y + \sqrt{x^2 + y^2} = cx^2$.

10. $\mathbf{2\frac{dy}{dx} = \frac{y}{x} + \frac{y^2}{x^2}}$

Sol^n: $2\frac{dy}{dx} = \frac{y}{x} + \frac{y^2}{x^2}$ ..... (i)

This equation is homogeneous differential equation

Let, $y = vx$ $\quad \therefore \frac{dy}{dx} = v + x\frac{dv}{dx}$

Then equation (i) will becomes

$$2\left(v + x\frac{dv}{dx}\right) = \frac{vx}{x} + \frac{v^2x^2}{x^2}$$

$\Rightarrow \quad 2v + 2x\frac{dv}{dx} = v + v^2 \qquad \Rightarrow 2x\frac{dv}{dx} = v^2 - v$

$\Rightarrow \quad 2x\frac{dv}{dx} = v(v-1) \qquad \Rightarrow \frac{dv}{v(v-1)} = \frac{1}{2}\frac{dx}{x}$

$\Rightarrow \quad \left(\frac{1}{v-1} - \frac{1}{v}\right)dv = \frac{1}{2}\frac{dx}{x}$

Integrating, we get

$$\log(v-1) - \log v = \frac{1}{2}\log x + \log c$$

$\Rightarrow \quad \log\frac{v-1}{v} = \log\sqrt{x}\,.\,c \qquad \Rightarrow \frac{v-1}{v} = \sqrt{x}\,c$

$\Rightarrow \quad v - 1 = \sqrt{x}\,.\,cv \qquad \Rightarrow \frac{y}{x} - 1 = \sqrt{x}\,.\,c\,\frac{y}{x}$

$\Rightarrow \quad y - x = \sqrt{x}\,cy \qquad \Rightarrow (y-x)^2 = cxy^2.$

11. $\mathbf{\frac{dy}{dx} = \frac{x-y}{x+y}}$**, y = 0 when x = 2**

Sol^n: $xdy + ydy = xdx - ydx$

$xdy + ydx = xdx - ydy$

$d(xy) = xdx - ydy$

Integrating, we get

$$\int d(xy) = \int xdx - \int ydx$$

$\Rightarrow \quad xy = \frac{x^2}{2} - \frac{y^2}{2} + c$

$\therefore \quad x = 2$, when $y = 0$

i. e. at (2, 0), $0 = 2 + c \qquad \Rightarrow c = -2$

$\therefore \quad xy = \frac{x^2}{2} - \frac{y^2}{2} - 2 \qquad \Rightarrow x^2 - 2xy - y^2 = 4.$

12. $\mathbf{(x^3 + y^3)dy - x^2\,y\,dx = 0,\ f(0) = 1}$

Sol^n: $\frac{dy}{dx} = \frac{x^2y}{x^3 + y^3}$ ..... (i), $f(0) = 1$

The equation being homogeneous differential equation.

Let $y = vx \quad \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

then equation (i) becomes,

$$v + x\frac{dv}{dx} = \frac{x^2\, vx}{x^3 + (vx)^3} \qquad \Rightarrow\ v + x\frac{dv}{dx} = \frac{v}{1 + v^3}$$

$$\Rightarrow\quad v\frac{dv}{dx} = \frac{v}{1 + v^3} = v \qquad \Rightarrow\ x\frac{dv}{dx} = \frac{v - v - v^4}{1 + v^3} = \frac{-v^4}{1 + v^3}$$

$$\Rightarrow\quad \frac{1 + v^3}{v^4}\,dv = -\frac{dx}{x} \qquad \Rightarrow \left(v^{-4} + \frac{1}{v}\right)dv + \frac{dx}{x} = 0$$

Integrating,

$$-\frac{1}{3v^3} + \log v + \log x = \log k.$$

$$\Rightarrow\quad \log v + \log x + \log k = \frac{1}{3v^2} \qquad \Rightarrow \log(xvk) = \frac{1}{3v^3} \ \ldots\ldots\ldots\ldots \text{(ii)}$$

$$\Rightarrow\quad \log\left(\frac{y}{x} \cdot x \cdot k\right) = \frac{1}{3\left(\frac{y}{x}\right)^3} \qquad \Rightarrow \log(ky) = \frac{x^3}{3y^3}$$

$$\Rightarrow\quad 3\log(ky) = \frac{x^3}{y^3} \qquad \Rightarrow 3\log k + 3\log y = \frac{x^3}{y^3}$$

$$\Rightarrow\quad \frac{x^3}{y^3} = 3\log y + c$$

at $x = 0,\ y = 1.0 = 3\log 1 + c \qquad \Rightarrow c = 0$

$\therefore\quad x^3 = 3y^3 \log y.$

Hints and solutions of MCQ's

1. The first order and first degree homogeneous differentiate equations is in the form of $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$
2. The differential equation $\frac{dy}{dx} = \frac{x + y - 1}{x + y + 1}$ is not expressible in the form $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$. So, it is not homogeneous differential equation.
3. The differential equation

$$\frac{dy}{dx} + \frac{y}{x} = \tan\left(\frac{y}{x}\right)$$

$$\Rightarrow\quad \frac{dy}{dx} = \tan\left(\frac{y}{x}\right) - \frac{y}{x} \text{ is of the form } \frac{dy}{dx} = f\left(\frac{y}{x}\right)$$

$\therefore$ it is homogeneous.

4. when $y = vx$

$$\frac{dy}{dx} = v + x\frac{dv}{dx} \qquad \Rightarrow \frac{dy}{dx} = f\left(\frac{y}{x}\right)$$

$$\Rightarrow\quad v + x\frac{dv}{dx} = f(v) \qquad \Rightarrow x\frac{dv}{dx} = f(v) - v$$

$$\Rightarrow\quad \frac{dx}{x} = \frac{dv}{f(v) - v}$$

Integrating,

$$\int \frac{dx}{x} = \int \frac{dv}{f(v) - v}$$

5. Let $\frac{y}{x} = v$

$\Rightarrow \quad y = vx$

$\Rightarrow \quad \frac{dy}{dx} = v + x\frac{dv}{dx}$

so, $\frac{dy}{dx} = \frac{y-x}{x} = \frac{y}{x} - 1$

$\Rightarrow \quad v + x\frac{dv}{dx} = v - 1 \qquad \Rightarrow x\frac{dv}{dx} = -1$

$\Rightarrow \quad dv = -\frac{dx}{x} \qquad \Rightarrow \int dv = -\int \frac{dx}{x}$

$\Rightarrow \quad v = -\log x + \log c \qquad \Rightarrow \frac{y}{x} = \log \frac{c}{x}$

$\Rightarrow \quad y = x \log \frac{c}{x}$

6. Let $\frac{y}{x} = v$

$\Rightarrow \quad y = vx$

$\Rightarrow \quad \frac{dy}{dx} = v + x\frac{dv}{dx}$

so, $\frac{dy}{dx} - \frac{y}{x} = \cos^2\frac{y}{x} \qquad \Rightarrow v + x\frac{dv}{dx} - v = \cos^2 v$

$\Rightarrow \quad x\frac{dv}{dx} = \cos^2 v \qquad \Rightarrow \sec^2 v dv = \frac{dx}{x}$

Integrating, we get

$\tan v = \log x + c \qquad \therefore \tan\frac{y}{x} = \log x + c$